Janmey Shah

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PROFILE

Biotechnology postgraduate with two years of lab experience in mammalian cell culture, cell-based assays, and GLP-compliant records. At ACTREC, maintained SH-SY5Y and U87 cells aseptically; expanded, passaged, counted and treated cells; and performed CCK-8 dose-escalation and Griess assays. Quantitative ELISA, Western blotting and cell-viability data established selectivity profiles across glioblastoma, SH-SY5Y and HMC3 microglial models. Skilled in structured assay and passaging logs, reagent preparation and laboratory readiness; contributed to data handovers, research discussions and presentations. Keen to apply this bench foundation to reliable cell culture and functional assay delivery, and to grow in routine assay optimisation and traceable workflows.

RESEARCH & LABORATORY EXPERIENCE

MSc Dissertation — Recombinant Protein Expression & Construct Engineering | *University of Leeds* | 2025 – 2026

- Performed Gibson Assembly cloning and recombinant protein expression of SurE in an NRPS expression system, designing and validating multi-fragment overlaps in SnapGene and optimising expression before SDS-PAGE verification
- Used SnapGene for construct and vector design and AlphaFold for structural analysis of NRPS assembly proteins and domains

Research Trainee — Cell Biology & Protein Science | *ACTREC, Mumbai, India* | Feb 2024 – Feb 2025

- Maintained three cell lines in triplicate sets and SH-SY5Y, U87, and HMC3 aseptically; managed adherent and suspension cultures, passage, banking, cryopreservation, and controlled thawing, with logs supporting consistent stocks, high viability, and zero contamination
- Ran CCK-8 dose-response assays with 2- to 5-fold serial dilutions; seeded 3,000–5,000 cells/well and read at 450 nm
- Mapped 96-well CCK-8 plates for 0–10 Gy irradiation; used 200 µL PBS in outer wells to reduce edge effects; internal wells held triplicate standards, vehicle/untreated controls, and radiation doses
- Ran Griess assays on supernatants; quantified nitric oxide against 7-point sodium nitrite standards (1.56–100 µM) by 2-fold serial dilution, achieving R^2 >0.99 linearity for quantitation
- Applied QC across triplicate/quadruplicate technical replicates with <5% CV; subtracted cell-free blanks, background-corrected, and normalized to vehicle controls for assay reproducibility
- Analyzed dose-response data by regression for IC50/survival fractions; completed 15+ independent in vitro radiobiology and cell-based studies of acute oxidative/inflammatory responses

Research Intern — Nanobiotechnology & Molecular Analysis | *Centre for Nanobiotechnology, VIT Vellore* | Aug 2022 – Aug 2023

- Conducted nanotoxicology experiments on material interactions using fluorescence microscopy and physiological markers, contributing to two Toxicology Reports publications
- Performed assays to quantify reactive oxygen species (ROS) generation and lipid peroxidation, providing data on oxidative-stress pathways in Chlorella sp.
- Managed reagent and consumable stocks, equipment maintenance, sample logistics, and shared laboratory organisation
- Experimental Execution: Executed standardized Algal Growth Inhibition Assays (OECD 201) to determine the EC-50 of Graphene Oxide and nanoplastic concentrations.

CORE SKILLS

Mammalian Cell Culture	Cell expansion, passaging, cryopreservation, aseptic technique, cancer and neuronal cell line maintenance
Molecular Biology	PCR, Gibson Assembly, DNA Cloning, SnapGene, Gel Electrophoresis, Western Blotting, Sanger Sequencing
Protein Expression	Recombinant protein expression, construct design and optimisation, SDS-PAGE quality assessment
Experimental Documentation	GLP-compliant records, accurate data entry, experimental reporting, SOP adherence
Lab Operations	Equipment maintenance, consumable stock management, sample preparation and handover coordination
Data & IT	Python, R, Excel, AlphaFold, PyMOL — data analysis, interpretation, and scientific reporting
Quality & Compliance	GCP-certified (NIHR), regulated research environment experience, attention to detail

EDUCATION

MSc — Biotechnology with Business Enterprise | *University of Leeds, UK* | Sep 2025 – Aug 2026

- Dissertation: Investigating protein-protein interactions in NRPS assembly systems, Gibson Assembly cloning, recombinant protein expression, SnapGene vector design, AlphaFold structural analysis
- Relevant Modules: Protein Engineering, Molecular Biology, Business of Biotech, Strategic Entrepreneurship

BTech — Biotechnology | *VIT Vellore, India* | Graduated 2024

- Grade: First Class Equivalent (GPA: 7.9/10)
- Core curriculum: Cell Biology, Molecular Biology, Biochemistry, Genetics, Protein Science

PUBLICATIONS

Contributing author on 4 peer-reviewed publications — evidence of careful, reproducible experimental work and high-quality scientific communication:

1. *"RAGE signaling pathway in glioblastoma and cognitive decline"* — Co-Author | **Cancer Letters** | Dec 2025
2. *"Pharmacological and therapeutic innovation to mitigate radiation-induced cognitive decline"* — CoAuthor | **Brain Research** | Jun 2025
3. *"Fluorescent nanoplastics increase toxic effects of Graphene oxide in Scenedesmus obliquus"* — Researcher | **Toxicology Reports** | Dec 2024
4. *"Diclofenac enhances Boron nitride nanoparticle toxicity in microalgae"* — Researcher | **Toxicology Reports** | Dec 2024

CERTIFICATIONS

- Introduction to Good Clinical Practice (GCP) | NIHR
- Biotech Research Students of India (BRSI) — Student Member